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# Configurable anti-ambipolar photoresponses for optoelectronic multi-valued logic gates

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## ABSTRACT

Anti-ambipolar transistors (AATs) are the leading platform for the paradigm shift from binary to multi-valued logic (MVL) circuits, increasing circuit integration density and data processing capacity. However, most AATs with p–n heterojunctions present limited controllability of the transconductance peak, which is key to MVL operation. Here, we report optically configurable AAT/bi-AAT photoresponses implemented with an InSe field-effect transistor for potential MVL operations. The charge trapping and detrapping processes incorporated with manually introduced trap states form the AAT peaks. Furthermore, leveraging a symmetric device configuration, the dark current is significantly suppressed, and AAT photoresponses are highlighted. Contributed by two pathways of trap states, the AAT/bi-AAT photoresponses are switchable by incident optical wavelength. This dependence facilitates optical wavelength to be one of the logic inputs for MVL, based on which we propose circuit-free ternary logic gates in a single device that can achieve more than ~6 and ~19 times improved data density (1 bit per transistor) for NMAX and XNOR, compared with such circuits in a traditional binary design. This work realizes optically controlled AAT photoresponses, paving the way to exploit optical wavelength as a new degree of freedom in MVL computing, offering a route toward ultra-high-density, ultra-low-power, and optically programmable optoelectronic integrated circuits.



The new era of hyper-Moore's Law demands increasing information density and data processing capacity within a given footprint beyond increasing the number of transistors through footprint reduction.[1] One representative approach to achieving this paradigm shift is the transcendence from binary to multi-valued logic (MVL) gates by allowing multiple bits in a single cell.[2–6] For the operation of MVL gates, anti-ambipolar transistors (AATs) require fewer devices to construct a single-bit cell with much less power consumption,[7–14] benefiting from negative differential transconductance.[15–19] Typical AATs are implemented with p–n heterojunctions by stacking various materials, exhibiting Λ-shaped transfer curves with a transconductance peak within a specific range of gate bias voltage ($V_{GS}$). This behavior is interpreted as the simultaneous occurrence of two electrical conduction pathways,[20–22] which is well-known to stem from the interlayer recombination of electrons and holes across the p–n heterointerface due to Shockley–Read–Hall and Langevin mechanisms arising from inelastic tunneling of majority carriers into trap states and Coulomb interaction, respectively.[20–22] However, despite recent advances in AATs (e.g., material combinations,[7–11] thickness modulation,[23,24] and phase[25] or strain[26] engineering), the all-electrical tuning of the transconductance peak limits the performance in optimizing MVL (e.g., it requires additional electrical terminals such as dual gates).[12–14] Instead, tunable transconductance AAT peaks via fully optical control by employing hybrid photoresponse mechanisms present a promising avenue to overcome these limitations.[27,28]

In this work, we report an optically configurable AAT photoresponses in an InSe field-effect transistor (FET). We leverage $CF_4$ plasma treatment to induce trap states in the InSe channel. The charge carriers can be trapped and detrapped, forming an AAT peak in the transfer curve. We further employ a symmetric device configuration

($CF_4$ plasma treated-untreated-treated) with $Si_3N_4$ passivation that protects the middle of the InSe channel. The created trap states shift the Fermi level at plasma treated regions, forming potential barriers that significantly suppress the dark current (<40 pA) and distinguish the AAT photoresponse with incident light (642 and 730 nm). Compared with traditional AAT devices (fully electrical operation with careful heterostructure selection), the AAT behavior in our configuration is triggered by optoelectronic processes via incident light. Furthermore, a bi-AAT photoresponse is observed with 532 nm light input, attributed to the wavelength-dependent carrier lifetime of extra trap states from the $Si_3N_4$ passivation layer. We tailor this optically configurable peak number to potential MVL operations of NMAX and XNOR. Our findings open a pathway to integrate multiple MVL gates in a single device without introducing complex processes or adding electrical terminals,[29–31] providing an ultimate strategy for optimizing circuit integration density and data processing capacity.

Charge trapping and detrapping are pivotal in nearly all modern electronic systems, such as floating-gate and charge-trap flash memories.[32] These trap-state-assisted processes are also reported for generating and electrically modulating AAT behaviors.[33–37] We chose InSe as a promising channel material[38–42] and use $CF_4$ plasma treatment to introduce trap states in the channel.[43] The InSe is mechanically exfoliated and transferred on a $SiO_2$/Si substrate. The flake is treated by $CF_4$ plasma for 4 min (~20 sccm under ~50 mTorr via RIE, Oxford Plasmalab 80Plus). After that two electrodes (5 nm Ti followed by 50 nm Au) are fabricated by e-beam lithography (Vistec EBPG 5000) and metal evaporation (MASA IM-9912).

The schematic of the InSe FET device is shown in Fig. 1(a). The InSe flake exhibits invisible change from the optical microscope images taken before and after fabrication [Fig. 1(b)]. This is further confirmed by the Raman characterization [Fig. S1(c), supplementary material], of which no new peak is observed. Energy-dispersive x-ray spectroscopy (EDX) spectra [Fig. 1(c)] indicates the presence of fluorine at the $CF_4$ plasma treated area. The EDX mapping [Fig. S1(d), supplementary material] of a reference device (treated by $CF_4$ plasma with a patterned mask) shows the vestigial fluorine. These results indicate that $CF_4$ plasma treatment can introduce fluorine to the InSe channel. The electrical characterization is implemented in a homebuilt setup. The transfer curve of the intrinsic InSe exhibit clear $n$-type behavior [Fig. 1(d)],[38,41] while its resistance increases significantly after treatment [Fig. 1(e)], resulting in a much lower level of source–drain current ($I_{DS}$). The AAT peaks are observed in the transfer curve both in forward gate sweep (from −200 to 200 V) and in backward gate sweep (from 200 to −200 V). The peak in forward sweep is much lower than that in backward sweep. This can be explained by the charge trapping and detrapping. The majority carriers (electrons) are injected into the channel from the source electrode and continuously drifted to the drain electrode under the source–drain voltage ($V_{DS}$) of 10 V. During the gate sweep, a switchable vertical electrical field is created. The direction of the electrical field depends on the applied gate voltage. A portion of the drifting electrons are trapped and detrapped by the trap states during the gate sweep. These detrapped electrons are collected by the drain electrode, forming a peak in the transfer curve [Fig. 1(f)]. Compared with conventional charge-trapping memory devices,[35–37] the trap states in our device are embedded within the InSe channel.



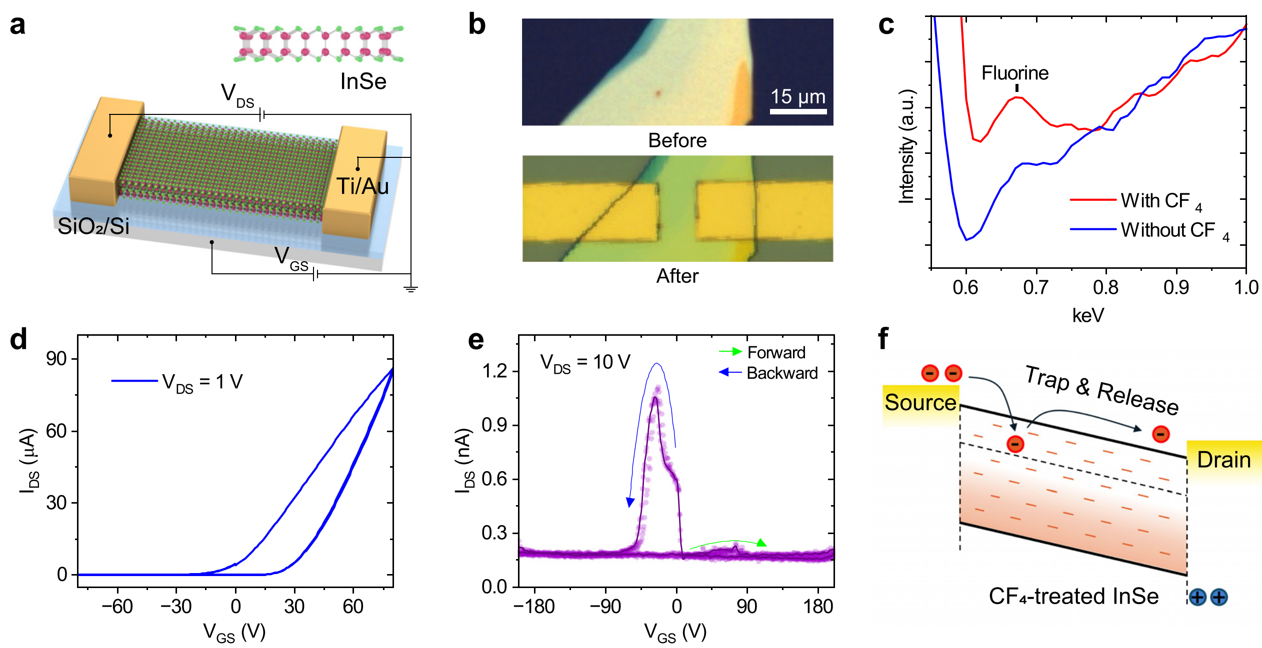


**FIG. 1.** $CF_4$ plasma induced trap states. (a) Schematic and (b) optical microscope image of the FET device before (top) and after (bottom) fabrication. (c) EDX spectra measured at $CF_4$ plasma treated (red) and untreated (blue) area. (d) and (e) The electrical transfer curve measured before (d) and after (e) $CF_4$ plasma treatment. After $CF_4$ plasma treatment, a clear AAT peak is observed. The transfer curve of intrinsic InSe in (d) is measured from a reference FET device. (f) The explanation of the AAT peak. The $CF_4$ plasma treatment introduces trap states into the InSe channel. The charge carriers in the channel get trapped and detrapped, forming a peak current on top of the baseline of $I_{DS}$. During the gate sweep, the number of contributed carriers contributes to the peak difference.

This feature enables charge trapping and detrapping during the backward sweep (see the supplementary material for details). Since the InSe exhibits *n*-type behavior, there are less contributed electrons at negative gate voltages, resulting in a lower peak in forward sweep (at ~60 V) and a higher peak in backward sweep (~−40 V).

We further employ a symmetric device configuration of treated-untreated-treated structure to suppress the AAT behavior in the dark condition [Fig. 1(e)] and achieve the optically controlled AAT photoresponse. Different from the previous device, a ~500 nm $Si_3N_4$ layer is deposited (PECVD, Oxford Plasmalab 80Plus) on top of the transferred InSe flake, serving as a passivation layer because $Si_3N_4$ has a broadband transparent window (from ultraviolet to mid-infrared) and can be etched through during $CF_4$ plasma treatment. The sample is then treated by $CF_4$ plasma for 14 min with a PMMA patterned mask. The etching speed for $Si_3N_4$ is ~50 nm/min, so that the InSe flake is treated for ~4 min, which is the same as the previous device. The patterned etching results in a $Si_3N_4$ passivation layer in the middle of the InSe channel. Finally, the electrodes are fabricated with the same tool, but an angled deposition (30°, 0°, −30°) is applied for a better connection.

The schematic of the InSe FET with $Si_3N_4$ passivation is shown in Fig. 2(a). Compared with the previous device, the InSe channel is treated with $CF_4$ plasma on two sides while remaining intrinsic in the middle region. The $Si_3N_4$ passivation is characterized to be ~527 nm, the InSe flake is ~110 nm, and the $CF_4$ plasma treated InSe is ~78 nm [Fig. 2(b)]. The photoluminescence peak of the InSe exhibits a red-shift from ~988 nm ($Si_3N_4$ passivated) to ~996 nm ($CF_4$ treated), and the peak intensity at $Si_3N_4$ passivated region is ~1.5 times higher than

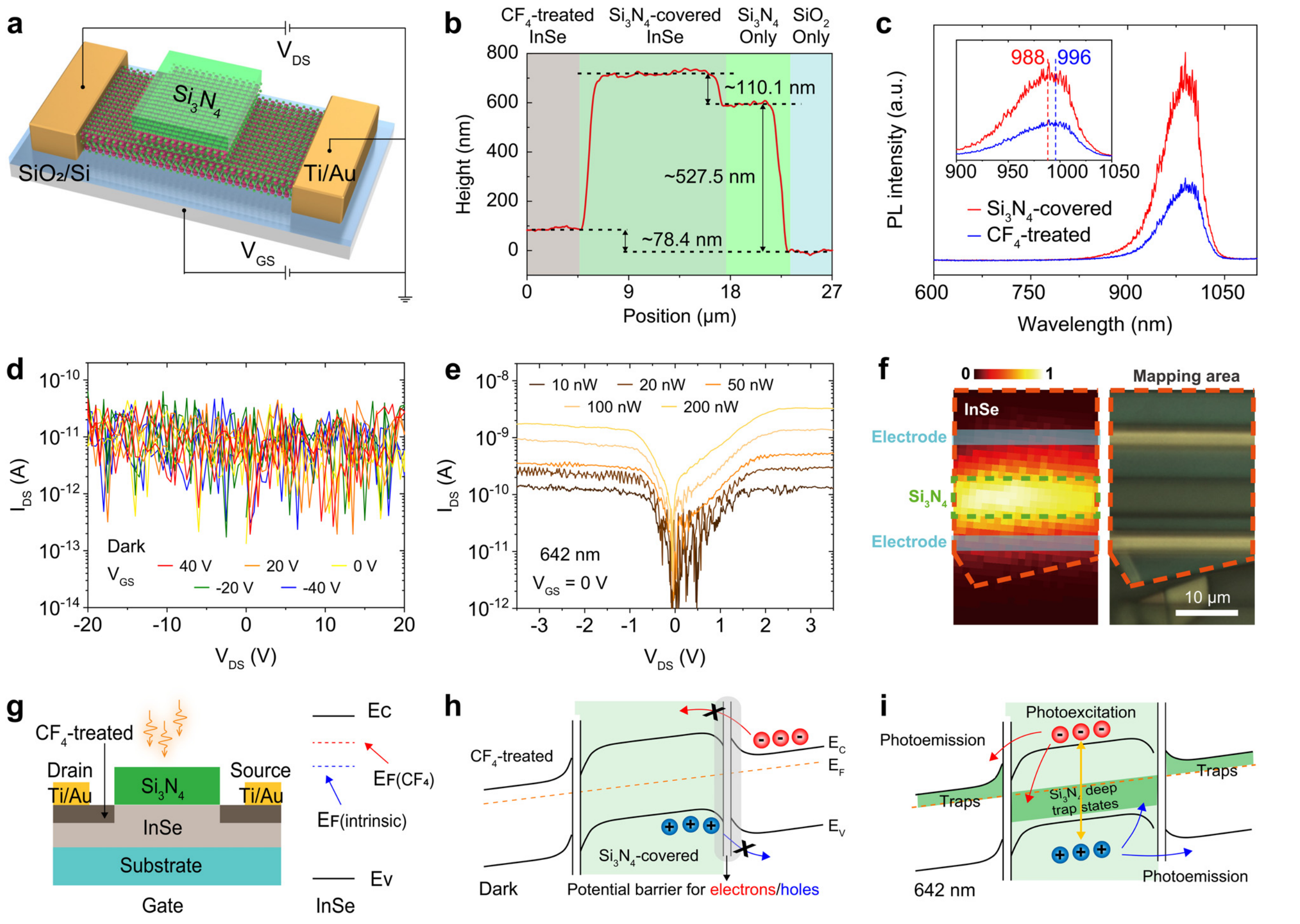


**FIG. 2.** Photoresponses of the symmetric device configuration with $Si_3N_4$ passivation. (a) Device schematic of the $CF_4$ plasma treated-untreated-treated symmetric structure with $Si_3N_4$ passivating middle region of the InSe channel. (b) Thickness characterization by AFM. The thickness of the InSe flake is calculated to be ~110 nm, and that of $Si_3N_4$ is ~527 nm. (c) Photoluminescence spectra measured at the $CF_4$ plasma treated (blue) and untreated (red) area. The $CF_4$ plasma treated area shows a peak red-shift from ~988 to ~996 nm. (d) Output curves measured in the dark condition with various $V_{GS}$ applied. In all conditions, the $I_{DS}$ remains below the system noise level (<40 pA). (e) Output curves measured from the device with different input light power. The input light is a 642 nm laser. (f) The scanning photocurrent mapping (left) and corresponding mapping area (right). The mapping is measured at $V_{DS} = 10$ V and $V_{GS} = 0$ V under light illumination of a 532 nm laser of 50 $\mu$ W. The orange and green dashed lines outline the InSe flake and $Si_3N_4$ layer, with the Ti/Au electrodes marked in cyan. (g) Schematic of the device with the potential barrier. The Fermi level of the $CF_4$ plasma treated InSe is shifted by the trap states. The presence of fluorine and the red-shift photoluminescence peak indicates a more *n*-type doped InSe. This shifted Fermi level results in a potential barrier at both side, forming a head-to-head diode structure that suppresses the dark current (h) and distinguishes the photocurrent (i).

the $CF_4$ plasma treated areas [Fig. 2(c)]. Different from the previous device, this InSe FET exhibits indiscernible $I_{DS}$ both in output curves [Fig. 2(d)] and in transfer curves [Fig. S3(a), supplementary material] in dark conditions. Instead, when the device is illuminated at the $Si_3N_4$ passivated area, a distinguishable $I_{DS}$ is measured, as plotted in Fig. 2(e) (and in Fig. S4). To visualize the distribution of the photocurrent, a scanning photocurrent mapping is implemented [Fig. 2(f)], from which the whole InSe channel exhibits photoresponse and the $Si_3N_4$ passivated area shows much higher signal. Considering the presence of fluorine and the deep levels (trap states) in the InSe channel, the Fermi level is assumed to be shifted upward (more $n$-type doped, and agreed with theoretical studies[44,45]) after the $CF_4$ plasma treatment [Fig. 2(g)]. Therefore, this device can be considered as a head-to-head diode structure. In dark conditions, the carrier transport is almost completely blocked by the anti-series potential barrier formed at the interface of $Si_3N_4$-passivated InSe and $CF_4$-treated InSe [Fig. 2(h)]. On the contrary, with sufficient energy (e.g., the photon energy for 642 nm laser is $\sim$1.93 eV) that exceeds the bandgap of InSe ($\sim$1.25 eV), the photoexcited carriers can surmount the potential barrier/well through photoemission [Fig. 2(i)], forming detectable photocurrent.

The transfer curves that exhibit AAT photoresponses are measured with 642 nm [Fig. 3(a)] and 730 nm [Fig. 3(b)] light input. The potential barrier/well across the channel significantly suppresses the dark current, while the photoemission and the charge trapping and detrapping processes contribute to the base current and AAT peaks of the transfer curve. The AAT photoresponses are measured down to a low incident power of even $\sim$10 nW at different wavelengths (Fig. S5). Interestingly, we observed a bi-AAT photoresponse when using 532 nm laser as input [Fig. 3(c)]. Different from the AAT photoresponses, a dual-peak transconductance appears during the gate sweep. This optical wavelength configurable AAT to bi-AAT transition can be explained by wavelength-dependent carrier detrapping processes[46,47] from the $CF_4$-treated InSe and the $Si_3N_4$ passivation layer [Fig. 3(d) and Fig. S6]. The photoexcited carriers possess the same energy as the excitation (e.g., $\sim$1.93 eV for 642 nm and $\sim$2.33 eV for 532 nm). These carriers are trapped both in the $CF_4$-treated InSe and the $Si_3N_4$ passivation layer.[48–50] These carriers are detrapped after a short period (lifetime)[43,46,47] and contribute to the photocurrent from different pathways [Fig. 3(d)]. The detrapping of these two pathways can be unsynchronized (considering the $Si_3N_4$ passivation layer is continuously illuminated, during which optically stimulated charge detrapping can happen[51]) resulting in a wavelength-dependent detrapping that causes the switchable AAT/bi-AAT photoresponses [Fig. 3(e)]. Furthermore, we find the competition of these two carrier pathways when changing the speed of gate sweep [Fig. 3(f)]. With the increase in speed, the peak tends to be indistinguishable and appears at lower absolute $V_{GS}$ values, which is attributed to the interplay of charge trapping and detrapping of these two pathways.

It is worth noting that the switchability from single to dual transconductance peaks featured in this work could not be achieved with electrical methods (e.g., by using dual gates).[7–14] In other studies,[23–27,52–56] switching from AAT to bi-AAT has been only reported under the application of external tensile strain[26] by mechanically inducing the artificial electric field to increase the rate of



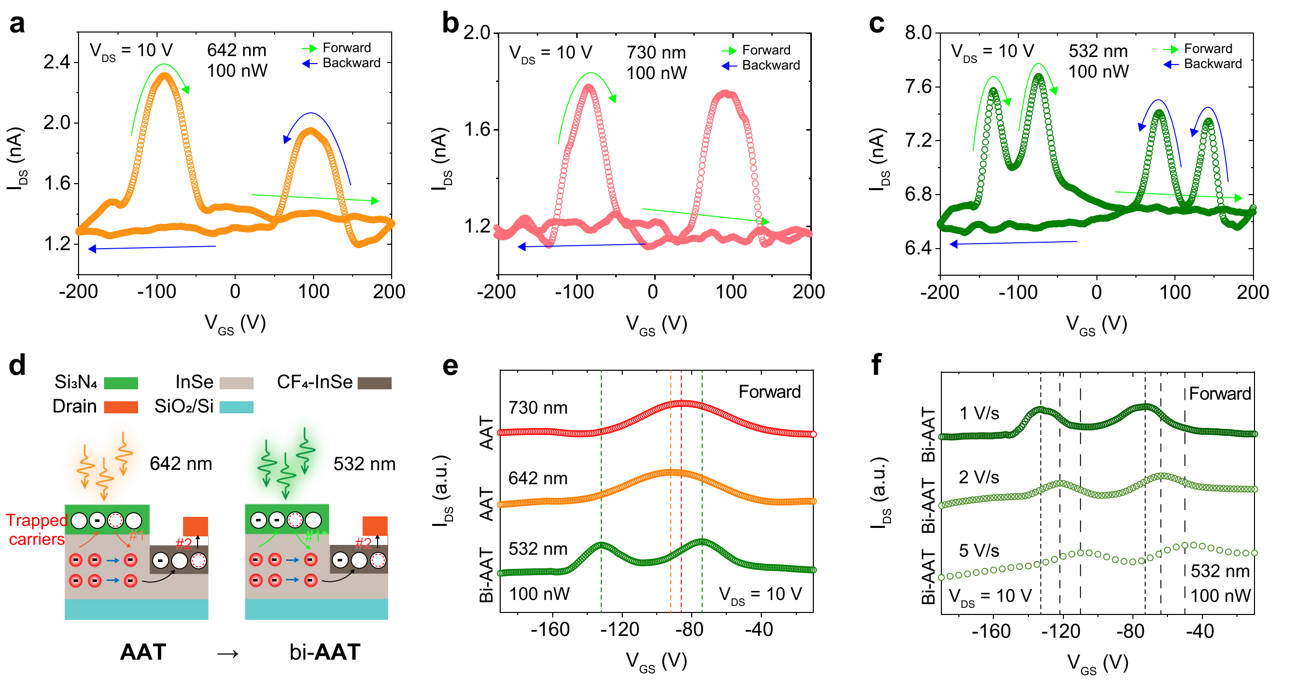


**FIG. 3.** AAT and bi-AAT photoresponse. (a)–(c) Transfer curves measured under light illumination at $V_{DS} = 10$ V. The light is (a) 642 nm, (b) 730 nm, and (c) 532 nm with a power of 100 nW. (d) Schematics explaining the formation of bi-AAT peaks. The presence of $Si_3N_4$ passivation layer provides extra trap states. The photoexcited carriers possess energy corresponding to the excitation wavelength. The competition of carrier detrapping from these two pathways leads to the wavelength-dependent AAT/bi-AAT photoresponses. (e) Normalized transfer curves at $V_{DS} = 10$ V in the forward $V_{GS}$ sweep under light illumination of 730 (upper), 642 (middle), and 532 (lower) nm with an incident power of 100 nW, and (f) measured under 532 nm light illumination with different $V_{GS}$ sweep speed of 1 (upper), 2 (middle), and 5 (lower) V/s.

interlayer recombination, forming an alternative carrier conduction path.[20–22] Compared with mechanical engineering, tuning the input wavelength is widely applicable to various platforms, which potentially meet the recent trend in electronic-photonic integrated circuits to boost the processing speed while reducing power dissipation in high bandwidth memory.

We further present a potential ternary logic gates by taking advantages of the switchability and tunability by optical wavelength featured in our observation. Due to the limited laser sources in our lab, we apply a simulation study for the concept demonstration. The wavelength-dependent transfer curves can be displayed as the waterfall [Fig. 4(a)] and heatmap [Fig. 4(b)] plots (normalized $I_{DS}$ as a function of light wavelength and $V_{GS}$) by using Synopsys HSPICE.[57] Various ternary gates can be implemented by exploring the variability of these plots. As representative examples, we introduce the ternary logic gates of NMAX (ternary NOR) [Fig. 4(c)] and XNOR (ternary exclusive NOR) [Fig. 4(f)] with a trinary digit of $I_{DS}$ controlled by the wavelength and $V_{GS}$. As verified through transient simulation [Figs. 4(d) and 4(g)], the three quantized logic states are defined by two current thresholds (0.73 and 0.86 for NMAX and 0.59 and 0.65 for XNOR) and denoted as −1 (green), 0 (magenta), and 1 (orange). For ternary logic gates, the state 0 is usually considered "unknown," which can be either −1 or 1. Therefore, in the NMAX gate, only inputs, which certainly lead to the "MAX" state ($-1 < 0 < 1$), will output the inverse state of the "MAX" state. Otherwise, the output remains 0 (unknown). Meanwhile, the XNOR gate outputs 1 for the same inputs and −1 for different inputs. On the other hand, 0 output only applies to 0 input cases. We further compared data density of NMAX and XNOR between our device in a circuit-free ternary design and a conventional circuit in a traditional binary design [Figs. 4(e) and 4(h)]. Surprisingly, the data unit that can be implemented per transistor for NMAX and XNOR has increased by ∼6 and ∼19 times, respectively. We note that NMAX is a universal gate that can build all other ternary logic gates with its combination, while XNOR is a 1-trit multiplier that can offer a typical arithmetic operation. Hence, our simulation for NMAX and XNOR suggest that higher-dimensional, complex, and multifunctional MVL operations are possible beyond the simple MVL logic gates simulated in other works, such as MVL NOT gates (inverters).[2–6] We expect that various multifunctional single-device MVL gates other than the ternary NMAX and XNOR gates demonstrated in this work

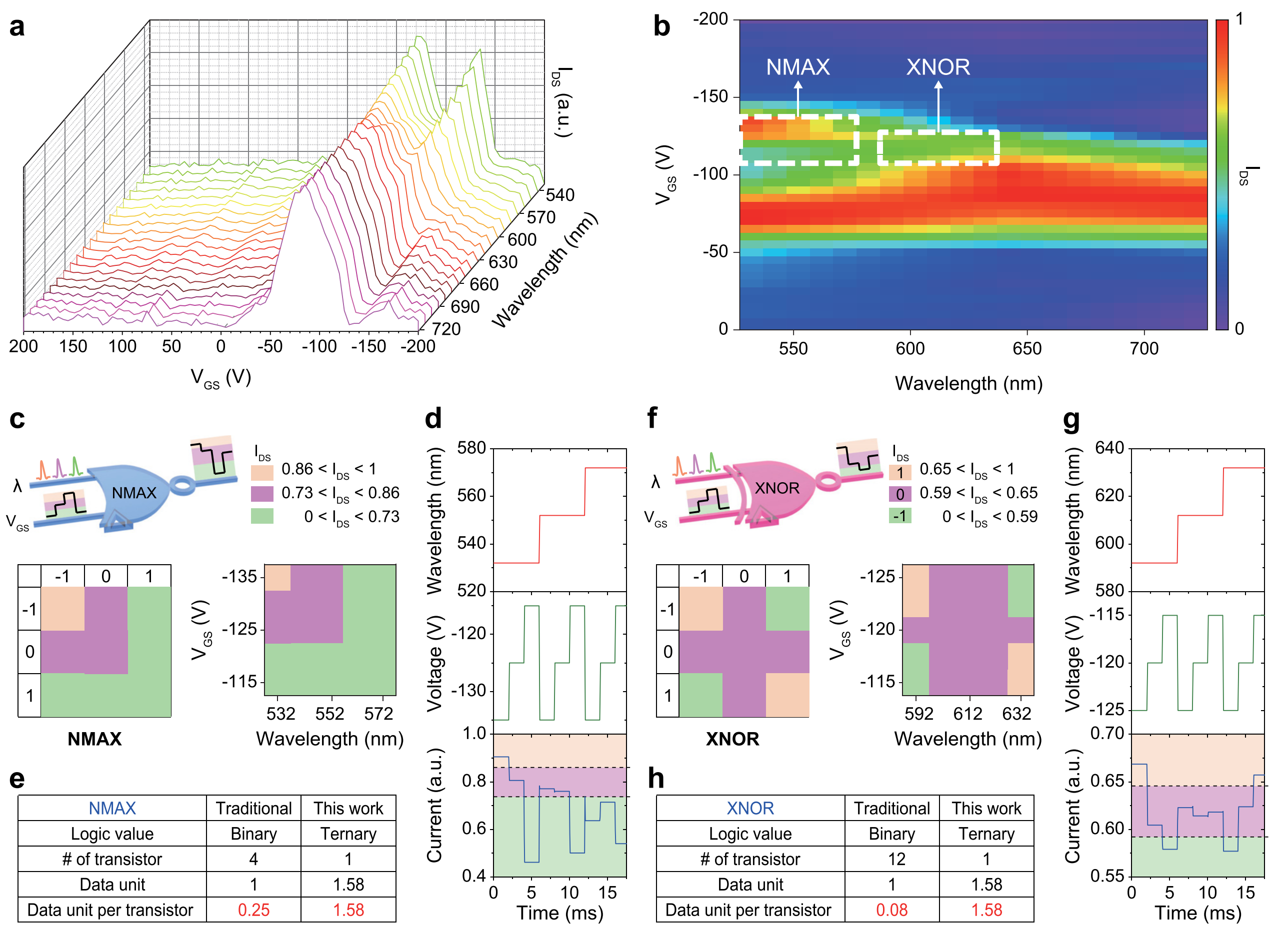


**FIG. 4.** Operation of ternary logic gates. (a) and (b) Wavelength-dependent transfer curves displayed as waterfall (a) and heatmap (b) plots. (c)–(h) Ternary logic gates of NMAX (c)–(e) and XNOR (f)–(h) with a trinary digit of $I_{DS}$ controlled by the light wavelength and $V_{GS}$. The truth tables in (c) and (f) are formularized based on the white dashed rectangles in (b) and verified through transient simulation (d) and (g). Data units that can be implemented per transistor are compared for NMAX (e) and XNOR (h).

will enable ultra-high-density and ultra-low-power MVL computing with simpler circuit architectures.[29–31]

In conclusion, we present an optically configurable AAT/bi-AAT photoresponses in the InSe FET with a $Si_3N_4$ passivation layer. Assisted by the symmetric device configuration and the barrier/well formed in the channel, we achieve significantly suppressed dark current and distinguishable AAT photoresponses. We further find the two pathways ($CF_4$-treated InSe and $Si_3N_4$) of photoexcited carriers contributed to a switchable AAT/bi-AAT photoresponses, configured by the input optical wavelength. This optically controlled AAT/bi-AAT photoresponses are tailored to realize MVL operations (NMAX and XNOR) in a single device with optimized area and energy efficiency. Our approach opens a new degree of freedom for optoelectronic MVL computing by utilizing optical wavelength as input.

See the supplementary material for the optoelectronic measurement, SPICE simulation, Raman characterization, AFM characterization, EDX mapping, photoswitching characteristics, wavelength- and power-dependent AAT photoresponse, charge trapping from the $Si_3N_4$ passivation layer, and the thresholds for a trinary digit.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Xiaoqi Cui:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Sunmean Kim:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Faisal Ahmed:** Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Mingde Du:** Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Andreas C. Liapis:** Formal analysis (equal); Methodology (equal); Writing – review & editing (equal). **Juan Arias Muñoz:** Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal); Writing – review & editing (equal). **Abde Mayeen Shafi:** Formal analysis (equal); Methodology (equal); Writing – review & editing (equal). **Md Gius Uddin:** Methodology (equal); Resources (equal); Writing – review & editing (equal). **Fida Ali:** Formal analysis (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **Yi Zhang:** Methodology (equal); Writing – review & editing (equal). **Dong-Ho Kang:** Resources (equal); Writing – review & editing (equal). **Harri Lipsanen:** Resources (equal); Writing – review & editing (equal). **Seokhyeong Kang:** Resources (equal); Writing – review & editing (equal). **Hoon Hahn Yoon:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zhipei Sun:** Funding acquisition (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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